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Flows, Magnetic Topology, and Temperature of a Transition Region Jet Observed by the EUV Snapshot Imaging Spectrograph

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ABSTRACT

Jets and explosive events in the solar transition region are thought to be driven by magnetic reconnection, but the spatial structure of their flows is difficult to observe, since a slit spectrograph records it one position at a time. The EUV Snapshot Imaging Spectrograph (ESIS) is a slitless, computed tomography imaging spectrograph which recorded the quiet Sun near disk center in O v 630 Å and several other transition region and coronal lines across a wide field of view during a sounding rocket flight on 2019 September 30. The largest event it observed, event E, is a jet with an inverted-Y morphology which lasted 14 min and extended 64 Mm. We invert the ESIS images into spectral cubes using the multiplicative algebraic reconstruction technique (MART), and measure Doppler shifts of up to **XX** toward and **XX** away from the observer. We compute the magnetic topology beneath the event from SDO/HMI magnetograms using magnetic charge topology, and compare the separatrices of the coronal field with the locations of the flows. We combine intensities from SDO/AIA, Hinode/XRT, and ESIS to compute differential emission measures of the event, and find a peak temperature of 2 MK. One or two sentences of conclusions: whether the flows are organized by the separatrices, and what the temperature implies about the coronal counterpart of the jet.

Keywords: Solar transition region (1532), Solar magnetic reconnection (1504), Solar magnetic fields (1503), Solar extreme ultraviolet emission (1493), Spectroscopy (1558), Doppler shift (401)

1. INTRODUCTION

The solar transition region (TR) is host to a variety of small, transient phenomena whose spectral line profiles reveal flows of order $100\,\mathrm{km\,s^{-1}}$. Explosive events, first observed by the High Resolution Telescope and Spectrograph (HRTS) as high-velocity jets in the quiet Sun (G. E. Brueckner & J.-D. F. Bartoe 1983), are compact, short-lived brightenings whose line profiles are broadened into one or both wings by Doppler shifts of about this size (K. P. Dere et al. 1989). D. E. Innes et al. (1997) found that the blue- and red-shifted emission of these events is separated along the slit, as expected of the bidirectional outflows from a reconnection site, and explosive events have since been widely interpreted as the signature of magnetic reconnection in the TR (K. P. Dere et al. 1991; D. E. Innes et al. 2015). They are associated with the evolution of the photospheric magnetic field (K. Muglach 2008), and a compact TR brightening has been found to occur in a magnetic field with a fan-spine topology (L. P. Chitta et al. 2017).

Jets are related, collimated phenomena which are observed across a wide range of temperatures, from network jets in the TR (H. Tian et al. 2014) to coronal X-ray jets (K. Shibata et al. 1992; N. E. Raouafi et al. 2016). Coronal jets commonly have an inverted-Y morphology, which is interpreted as reconnection between a small closed bipole and the surrounding large-scale field (K. Shibata et al. 1992; N. E. Raouafi et al. 2016), at a null point whose fan surface divides the two flux systems (E. Pariat et al. 2009). They are divided into standard and blowout jets (R. L. Moore et al. 2010), and many have been found to be driven by the eruption of a minifilament, in coronal holes (A. C. Sterling et al. 2015), in active regions (A. C. Sterling et al. 2016), and in the quiet Sun, where the eruption is triggered by flux cancellation (N. K. Panesar et al. 2016). The breakout model unifies these observations as reconnection at a null point above an erupting filament channel (P. F. Wyper et al. 2017, 2018). Whether explosive events, network jets, and coronal jets are the same process at different

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scales remains open (Y. Chen et al. 2019), and one part of that question is whether explosive events have a coronal counterpart at all (L. Teriaca et al. 2002).

These models make specific predictions about where flows should appear relative to the magnetic structure, and testing them requires the spectrum at every point of an event at the same time. A slit spectrograph samples only one position at a time, so a raster across an evolving event conflates its spatial and temporal structure, while an imager records the morphology of the event but measures flows only in the plane of the sky. The EUV Snapshot Imaging Spectrograph (ESIS) was designed to resolve this ambiguity. It is a slitless spectrograph which records several dispersed images of the Sun at once, each dispersed in a different direction, from which a spectral cube of the whole field can be recovered in the manner of a computed tomography imaging spectrograph (CTIS) (T. Okamoto & I. Yamaguchi 1991; M. Descour & E. Dereniak 1995). It is the successor to the Multi-Order Solar EUV Spectrograph (MOSES) (C. C. Kankelborg & R. J. Thomas 2001), whose flights recorded explosive events in He II 304 Å (J. L. Fox et al. 2010; T. Rust & C. C. Kankelborg 2019) and showed how the spectral content of overlapping images can be recovered (H. T. Courrier & C. C. Kankelborg 2018; J. D. Parker & C. C. Kankelborg 2022).

ESIS flew for the first time on 2019 September 30. J. D. Parker et al. (2022) reported the first results of that flight, including the Doppler shifts of several explosive events in O V 630 Å, and identified among them event E, the largest and most complex velocity event the instrument observed. In the images of event E they found a strong redshift at its brightest point and faint, blue-shifted material rising from it along a thin structure which is also visible in the Atmospheric Imaging Assembly (AIA) images, which they suggested might be a small jet or minifilament eruption (A. C. Sterling et al. 2015). Its complexity defeated an interpretation from the images alone, and they deferred its analysis to a future publication. Here we recover its spectra by inverting the ESIS images with the multiplicative algebraic reconstruction technique (MART) (R. Gordon et al. 1970; T. Okamoto & I. Yamaguchi 1991), which reconstructs the spectral cube at every exposure.

If event E is driven by reconnection, its flows should originate where reconnection can occur. In magnetic charge topology (MCT) (D. W. Longcope & I. Klapper 2002; D. W. Longcope 2005), the photospheric field is represented by discrete magnetic sources, and the potential coronal field above them is divided into domains of distinct connectivity by separatrix surfaces, which intersect along separators where reconnection transfers flux between domains. MCT has been applied to active regions (G. Barnes et al. 2005), and the X-ray brightness of coronal bright points is consistent with the energy released by reconnection at the separator between a converging pair of sources (D. W. Longcope et al. 2001). Computed from the Helioseismic and Magnetic Imager (HMI) magnetograms (P. H. Scherrer et al. 2012) beneath event E, the separatrices predict where its bidirectional flows should be, and the Doppler maps from ESIS show where they are.

A second question is how hot the event becomes. Differential emission measures (DEMs) are routinely computed from the extreme ultraviolet (EUV) channels of AIA (J. R. Lemen et al. 2012) using a variety of methods (I. G. Hannah & E. P. Kontar 2012; J. Plowman et al. 2013; M. C. M. Cheung et al. 2015; J. Plowman & A. Caspi 2020), but those channels are dominated by coronal lines and constrain the TR poorly, while the X-Ray Telescope (XRT) (L. Golub et al. 2007) is sensitive to the hottest plasma. The ESIS passband contains lines of He I, O III, O IV, O V, and Mg X, which together span the TR and the low corona, so that combining all three instruments constrains the DEM of the event from the TR to the corona.

In this paper we analyze event E with all three of these tools. Section 2 describes the observations from ESIS, AIA, HMI, and XRT and gives an overview of the event. Section 3 describes the inversion of the ESIS images and the Doppler shifts of the event, Section 4 the magnetic topology computed from HMI, and Section 5 the DEMs. We discuss what they imply about the mechanism of the event in Section 6 and conclude in Section 7.

2. OBSERVATIONS

Outline.

- **The ESIS flight** (J. D. Parker et al. 2022). Launched from White Sands Missile Range at 18:04 UT on 2019 September 30, with 10 s exposures from 18:06:11 to 18:11:01 UT. The octagonal field of view (FOV), about 11.5′ across, was pointed near disk center during an exceptionally quiet period, in which GOES had detected no flare of B class or above since 2019 July 7. The passband and its lines: He I 584 Å, O III 600 Å, O IV 608 Å and 610 Å, Mg X 610 Å and 625 Å, and O V 630 Å, with the temperature at which each is formed.
- **Coordinated observations.** AIA EUV images at a 12 s cadence (J. R. Lemen et al. 2012); HMI LOS magnetograms at a 45 s cadence, about seven of which cover the flight (P. H. Scherrer et al. 2012); XRT images through its Al_poly filter, the only X-ray filter it used during the flight, of a field 6.6′ across near disk center at 1.03″ per pixel, at a cadence of about 18.5 s throughout the flight (L. Golub et al. 2007; T. Kosugi et al. 2007). The Interface Region Imaging Spectrograph (IRIS) ran coarse rasters near disk center during

the launch window, but its slit never crossed event E, so there are no IRIS spectra of it; its 1400 Å slit-jaw images show most of event E, which is cut off by the edge of their field.

- **Co-alignment.** ESIS to AIA 304 Å (which correlates best with He I), and HMI and XRT to AIA, with the residual misalignment that bounds the comparisons in Sections 4 and 5.
- **Overview of event E.** Its location, (47.8″, −87.8″), about 100″ from disk center, and its lifetime (14 min, from 18:02:54 to 18:16:30 UT), including the parts before and after the ESIS exposures from AIA, over which its base is brighter in AIA 304 Å, the channel in which it brightens first, than before the event by more than 10 % of its peak brightening. Its inverted-Y morphology in O V and in the AIA channels (Figure 2), an order of magnitude brighter than the surrounding network, and the northward drift of its blue-shifted side by about 11″ over four minutes. Its length (64 Mm), in the plane of the sky from its base to its northern footpoint, where the fainter arc which continues its spire ends. Its light curves in every EUV channel of AIA and in XRT (Figure 1), from which its lifetime is measured.
- **Phases of the event**, as seen in the preliminary inversions: (1) blue shifts of about 60 km s^{-1} directly beside strong red shifts near the right footpoint of the inverted Y; (2) the blue-shifted component migrates to the cusp while the red-shifted component stays at the footpoint; (3) a blue-shifted blob moves along the right leg at an apparent speed of 100 km s^{-1} to 200 km s^{-1} in the plane of the sky.
- **Figures:** the full ESIS field with event E boxed; the light curves of event E (Figure 1); a time series of event E in ESIS O V, AIA 304 Å, 171 Å and 193 Å, XRT, and HMI (Figure 2).

3. SPECTRAL INVERSION AND DOPPLER SHIFTS

Outline.

- **Forward model.** Each ESIS channel records a dispersed image of the field, so the images are a linear function of the spectral radiance $I(x, y, \lambda)$. Describe the model of that function built from the flight optics (the `esis` and `optika` packages), including the point-spread function and the vignetting.
- **The MART inversion** (R. Gordon et al. 1970; T. Okamoto & I. Yamaguchi 1991), implemented in the `ctis` package. The update rule, the initial guess, and the stopping criterion. The reconstructed cubes sample the sky every 0.75″ and the line of sight every 17.5 km s^{-1} over ±210 km s^{-1} about each line, in five spectral windows, and are reconstructed independently at each exposure. How the blend of O IV and Mg X near 610 Å is handled, and why the radiometry of the channels is relative, so that velocities are trusted further than intensities.

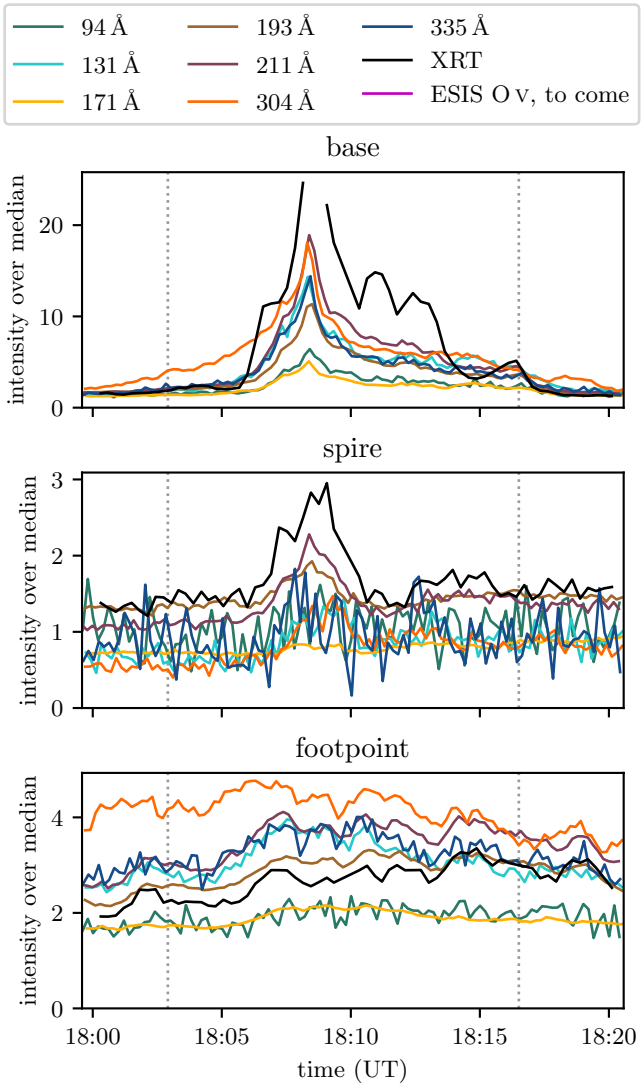


Figure 1. The light curves of the base of event E (top), the hottest part of its spire (middle), and its northern footpoint (bottom), the three points circled in Figure 5, in each EUV channel of AIA, labeled by its wavelength, and in XRT Al_poly, at the middle of every exposure from 17:59:35 to 18:20:34 UT. Each is the mean intensity of the pixels within 3″ of the point, or within 1.5″ of the spire, which is narrower, over the median intensity of its image, so that 1 is the median in every channel. The curve of XRT at the base has a gap where XRT saturated. The dotted lines mark the lifetime of event E, from 18:02:54 to 18:16:30 UT, over which the base is brighter in AIA 304 Å than in the first 2 min of the curve by more than 10 % of its peak brightening. The ESIS O V curves are left out until the ESIS data are ready.

- **Validation.** Invert synthetic ESIS images of a known scene containing a bidirectional jet, and report how well the recovered Doppler shifts and widths match the truth as a function of brightness and velocity. This sets the uncertainties quoted for event E, and the speed

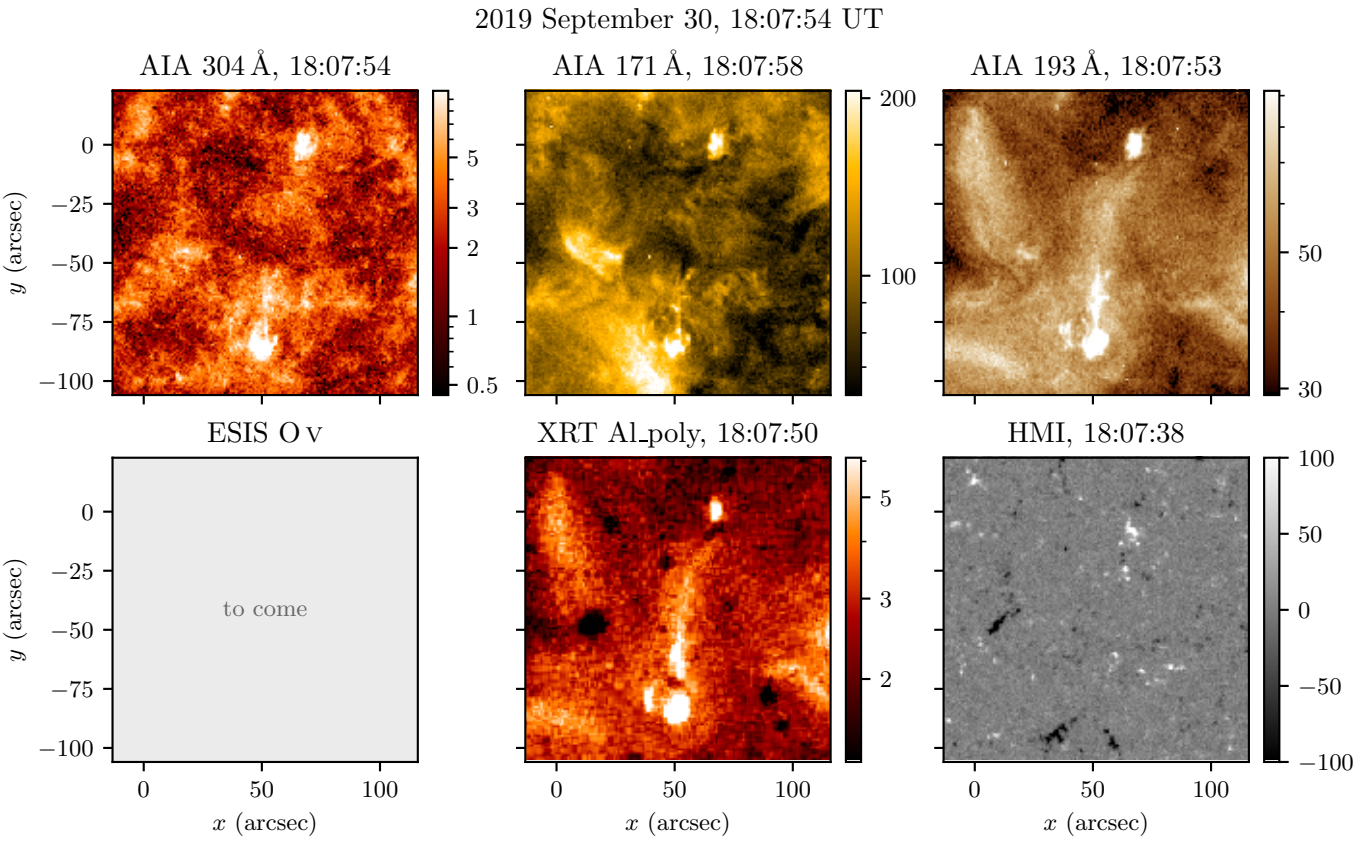


Figure 2. Event E in ESIS O v, in AIA 304 Å, 171 Å and 193 Å, in XRT Al_poly, and in the LOS magnetograms of HMI at 18:07:54 UT, close to the time of the still of Figure 5, over its field. Each panel shows the image of its instrument whose exposure is centered closest to that time, on its own pixels, and is labeled with the middle of that exposure. The colorbars give the intensity of AIA and XRT in DN s^{-1}, and the LOS field of HMI in gauss. The brightness of each channel is on a logarithmic scale between percentiles 1 and 99.5 of its intensity over the field in every frame of the animation, and the magnetograms saturate at ±100 G, white where the field points toward the observer and black where it points away. The animation shows each of the 105 images of AIA 304 Å from 17:59:42 to 18:20:30 UT, beside the image of each other instrument closest to it in time, at 4 frames per second, and runs for 26 s. The panel of ESIS is empty outside of its exposures, from 18:06:11 to 18:11:11 UT. The ESIS O v images are left out until the ESIS data are ready. The animation can be watched online.

beyond which a fast component is poorly constrained by the spectral window.

- **Doppler shifts of event E.** Line-of-sight velocity and width maps from the inverted cubes, from moments and from fits of two Gaussian components to each O v profile, a core near rest and a fast component. Peak speeds of the blue and red components (**XX** and **XX**). Preliminary fits find blue wings reaching about 150 km s^{-1} and a separate red component near 100 km s^{-1}, while the median Doppler shift, which moves only part of the way toward a fast component in one wing, peaks near 60 km s^{-1}; settle which of these the article quotes.

- **Other lines.** The red-shifted footpoint appears in O III and O IV, while the blue-shifted material at the cusp is seen only in O v and faintly in He I; whether this is a matter of brightness or of temperature.

- **Motion.** The plane-of-sky speed of the blue-shifted component, combined with its line-of-sight speed into a three-dimensional velocity.

- **Figures:** the inverted line profiles at a few points in the event; a time series of Doppler maps of event E; the plane-of-sky track of the fast components.

4. MAGNETIC TOPOLOGY

Outline.

- **Magnetograms.** The HMI LOS magnetograms (P. H. Scherrer et al. 2012) around the time of the flight, their co-alignment with ESIS, and the evolution of the photospheric flux beneath event E over the preceding hours: is there a parasitic polarity at the footpoint of the inverted Y, and is it emerging or cancelling (N. K. Panesar et al. 2016)?

- **MCT** (D. W. Longcope & I. Klapper 2002; D. W. Longcope 2005). Partition the magnetogram into unipolar sources, represent each as a point charge, and compute the potential field they produce (G. Barnes et al. 2005). Locate its null points, spines, separatrix surfaces, and separators, and in particular any fan-spine structure above the event and the height of its null.
- **Separatrices and flows.** Overlay the separatrix footprints and the location of any null point or separator on the Doppler maps of Section 3. Test whether the blue and red components lie on opposite sides of a separatrix, whether the cusp of the inverted Y lies beneath the null, and whether the migration of the blue-shifted component to the cusp follows a separatrix.
- **Energetics.** The flux transferred across the separator during the event, and the energy that its reconnection could release (D. W. Longcope et al. 2001).
- **Robustness.** How the topology changes with the partitioning threshold and from one magnetogram to the next, and whether the features that matter to event E persist in the weak fields of the quiet Sun.
- **Figures:** the magnetogram with its sources and separatrix footprints; the same footprints overlaid on the Doppler maps; a three-dimensional view of the separatrices and the null.

5. TEMPERATURE

Outline.

- **Data.** AIA EUV intensities (J. R. Lemen et al. 2012), XRT intensities (L. Golub et al. 2007), and the ESIS line intensities, co-aligned and averaged over the event and over a background region of network, at several times through the event.
- **Temperature responses.** The AIA and XRT responses and the contribution functions of the ESIS lines, all computed from CHIANTI (G. Del Zanna et al. 2021) with the same abundances and ionization equilibrium so that the instruments can be combined. The ESIS lines of O III, O IV, and O V extend the coverage of AIA and XRT down into the TR, and Mg X overlaps the coolest coronal channels of AIA. He I and AIA 304 Å are optically thick and are left out of the inversion. The AIA and XRT responses the DEMs are inverted from so far are plotted in Figure 3.
- **Radiometric calibration.** The ESIS intensities are needed in absolute units here, unlike in Section 3; how the channels are cross-calibrated against AIA and the uncertainty this contributes.
- **DEM inversion** (J. Plowman et al. 2013; J. Plowman & A. Caspi 2020), implemented in the `utu` package, cross-checked against an independent method (I. G. Hannah & E. P. Kontar 2012; M. C. M. Cheung et al. 2015). Report the DEMs with and without each instrument, to show what each contributes (Figure 4).

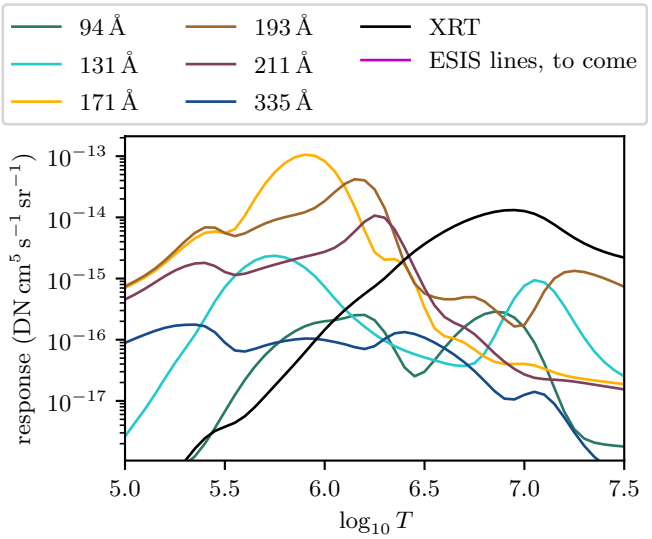


Figure 3. The temperature responses from which the DEM of event E is inverted, per unit emission measure and per steradian, from $\log_{10} T = 5.0$ to 7.5: each EUV channel of AIA, labeled by its wavelength, as computed by SolarSoft (P. Boerner et al. 2012), normalized to the EUV Variability Experiment (EVE) (T. N. Woods et al. 2012) and corrected for the emission missing from CHIANTI, and XRT Al_poly, as computed by xrtpy (N. Narukage et al. 2011), with the contamination on the date of the flight. AIA 304 Å is left out, since He II is optically thick. The ESIS lines of O III, O IV, O V, and Mg X are left out until the ESIS line list is ready. The AIA and XRT responses come from different versions of CHIANTI, where the outline calls for one.

- **Results.** The DEM of event E compared with that of its surroundings, its peak temperature (2 MK, where the DEM of the base from AIA and XRT in Figure 4 peaks), and whether the event has a coronal component visible to XRT or is confined to the TR, the question raised by L. Teriaca et al. (2002). The DEM of the base also has a second, cooler peak in the TR, which AIA and XRT constrain poorly and the ESIS lines will pin down. Whether the temperature differs between the cusp and the footpoint of the inverted Y.
- **Evolution.** The DEM of event E from AIA and XRT for every XRT image from 18:00:12 to 18:19:58 UT (Figure 5), inverted as in the DEM tutorial of the `hinode` package, at the full resolution of XRT, with and without the background subtracted.
- **Figures:** the temperature responses of the three instruments (Figure 3); the DEMs of event E and of the background (Figure 4); the DEM of event E through its lifetime (Figure 5).

6. DISCUSSION

Outline.

- **A reconnection jet?** Bring together the flows, the topology, and the temperature into a single picture

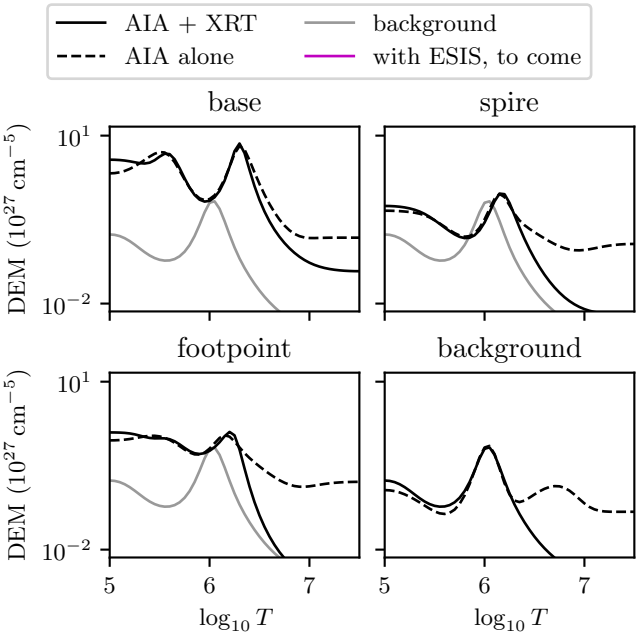


Figure 4. The DEMs of the base of event E, the hottest part of its spire, its northern footpoint, and a background region of network at 18:08:00 UT, the time of the still of Figure 5, each inverted from the mean intensities of the pixels within 3″ of the point, 1.5″ for the spire, or within 5″ of (95″, −10″) for the background, from the six EUV channels of AIA alone (dashed) and with XRT Al_poly (solid). The DEM of the background from AIA and XRT is repeated in each panel (gray). The DEMs from AIA alone predict 1.6 to 8.6 times the XRT intensity observed, since AIA alone leaves room for a tail of hot plasma which XRT rules out, and those with XRT predict it to within 9 %. The DEMs with the ESIS lines are left out until the ESIS line list is ready.

of event E, and ask whether it is consistent with reconnection at a null point or separator driving bidirectional outflows (D. E. Innes et al. 1997; E. Pariat et al. 2009).

- **A minifilament eruption?** Compare the three phases of the event with the minifilament eruption picture (A. C. Sterling et al. 2015, 2016): adjacent blue and red jets as the rising minifilament reconnects internally, the blue component moving to the cusp as it reconnects with the surrounding field, and the blob of the third phase as a plasmoid. Compare the size of the dome, the height of the null, and the speed and duration of the jet with the breakout model (P. F. Wyper et al. 2017, 2018), and decide between standard and blowout (R. L. Moore et al. 2010).
- **Energetics.** The kinetic and thermal energy of the event compared with the magnetic energy that the topology suggests was released.
- **Context.** Where event E falls among TR explosive events (G. E. Brueckner & J.-D. F. Bartoe 1983; K. P. Dere et al. 1989), network jets (H. Tian et al. 2014; Y. Chen et al. 2019), and coronal jets (K. Shibata et al. 1992; N. E. Raouafi et al. 2016), and what it says about whether explosive events have a coronal counterpart (L. Teriaca et al. 2002).
- **Snapshot spectroscopy.** What a slit raster would have recorded of the same event, and which conclusions depended on observing the whole event at once.

7. CONCLUSIONS

Outline.

- Summarize the measured flows, topology, and temperature of event E.
- State the answer to the question posed in the introduction: whether the bidirectional flows of event E are organized by the separatrices of the coronal field.
- Implications for future snapshot spectrographs, including the second flight of ESIS.

ACKNOWLEDGMENTS

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APPENDIX

A. BACKGROUND SUBTRACTION

For our collaborators, to compare two sizes of the background; probably to be removed before submission. The bottom row of Figure 5 is inverted from the intensities of each pixel less their background, the trimmed mean of the 61 by 61 pixels of XRT (63″) around it. Figures 6 and 7 compare that background, at the time of the still, with the trimmed mean of the 31 by 31 pixels (32″) around each pixel. The wider background is smoother, and leaves more of

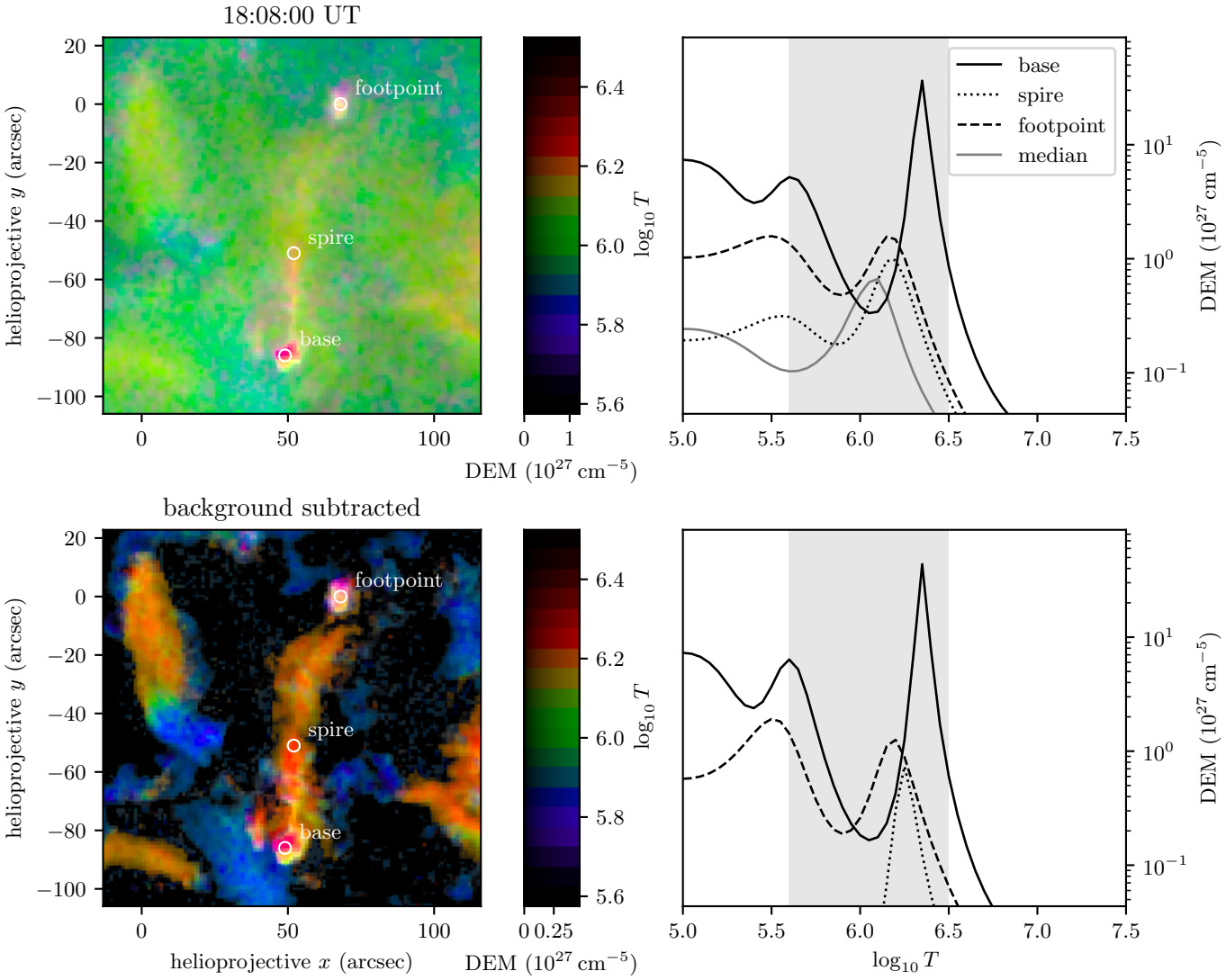


Figure 5. The DEM of event E from the six EUV channels of AIA and from XRT Al_poly at 18:08:00 UT, as a false-color image (left), inverted from the intensities as observed (top) and with the background of each pixel subtracted first (bottom): in each channel, the mean of the 61 by 61 pixels of XRT (63″) around it, without the brightest and the faintest 25 % of them, which leaves out event E. Each pixel is colored as if the DEM were the spectrum of the light it emits, with $\log_{10} T$ from 5.6 to 6.5 mapped onto the visible spectrum, from violet for the coolest plasma to red for the hottest (middle). Every temperature and every frame of a row shares one linear scale, up to the 99.5th percentile of the row over the whole movie, so the brightness of a pixel is how much plasma it has and its color is at what temperature. White pixels are those where XRT saturated, which have no DEM, and black pixels in the bottom row are those no brighter than their background in any channel. The DEM of the pixel nearest each of the three circled points, the base of event E, the hottest part of its spire, and its northern footpoint, is plotted over every temperature of the inversion (right), against the median DEM of the pixels in the image (gray line, top only, since the median pixel is no brighter than its background), with the temperatures colored in the image shaded gray. The animation shows the DEM for each of the 64 XRT images which began from 18:00:12 to 18:19:58 UT, at 4 frames per second, and runs for 16 s. The animation can be watched online.

the loops around event E behind when it is subtracted. Under the base of event E, either background is a small part of the intensity in every channel. Under the spire, either is most of it, and its DEM is close to that of the spire itself, so the DEM of the spire with the background subtracted is the one which depends the most on how the background is chosen.

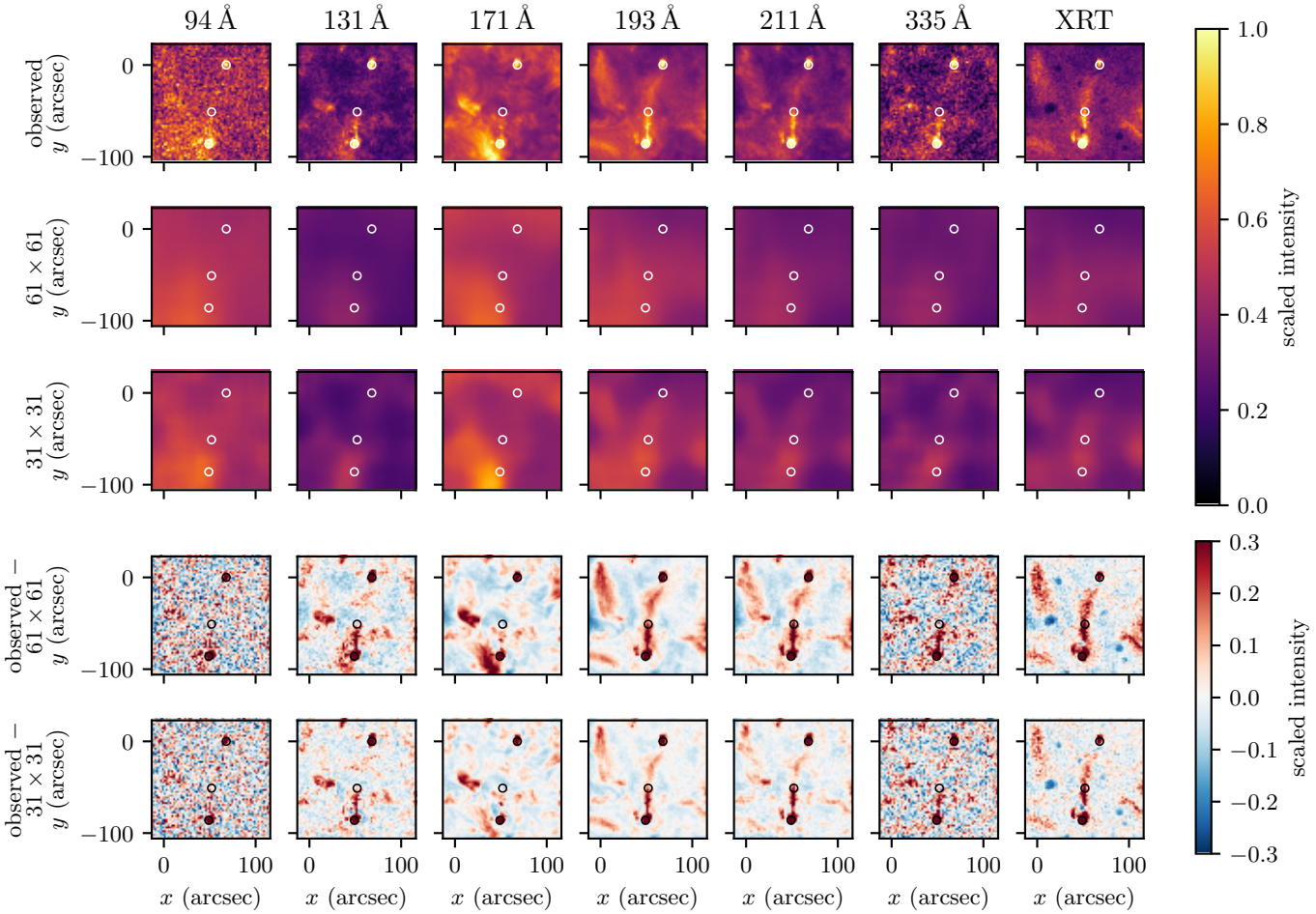


Figure 6. The intensities of event E at 18:08:00 UT, the time of the still of Figure 5, in each EUV channel of AIA, labeled by its wavelength, and in XRT Al_poly (top row), on the pixels of XRT from which the DEM is inverted. Beneath them are two backgrounds, the mean of the 61 by 61 pixels (63″) around each pixel, which the bottom row of Figure 5 subtracts, and that of the 31 by 31 pixels (32″) around it, each without the brightest and the faintest 25 % of them, and beneath those, what is left when each is subtracted (bottom two rows, red where the intensity is above the background and blue where it is below). Each channel is scaled to the 99.5th percentile of the intensity observed in it. The base of event E, the hottest part of its spire, and its northern footpoint are circled.

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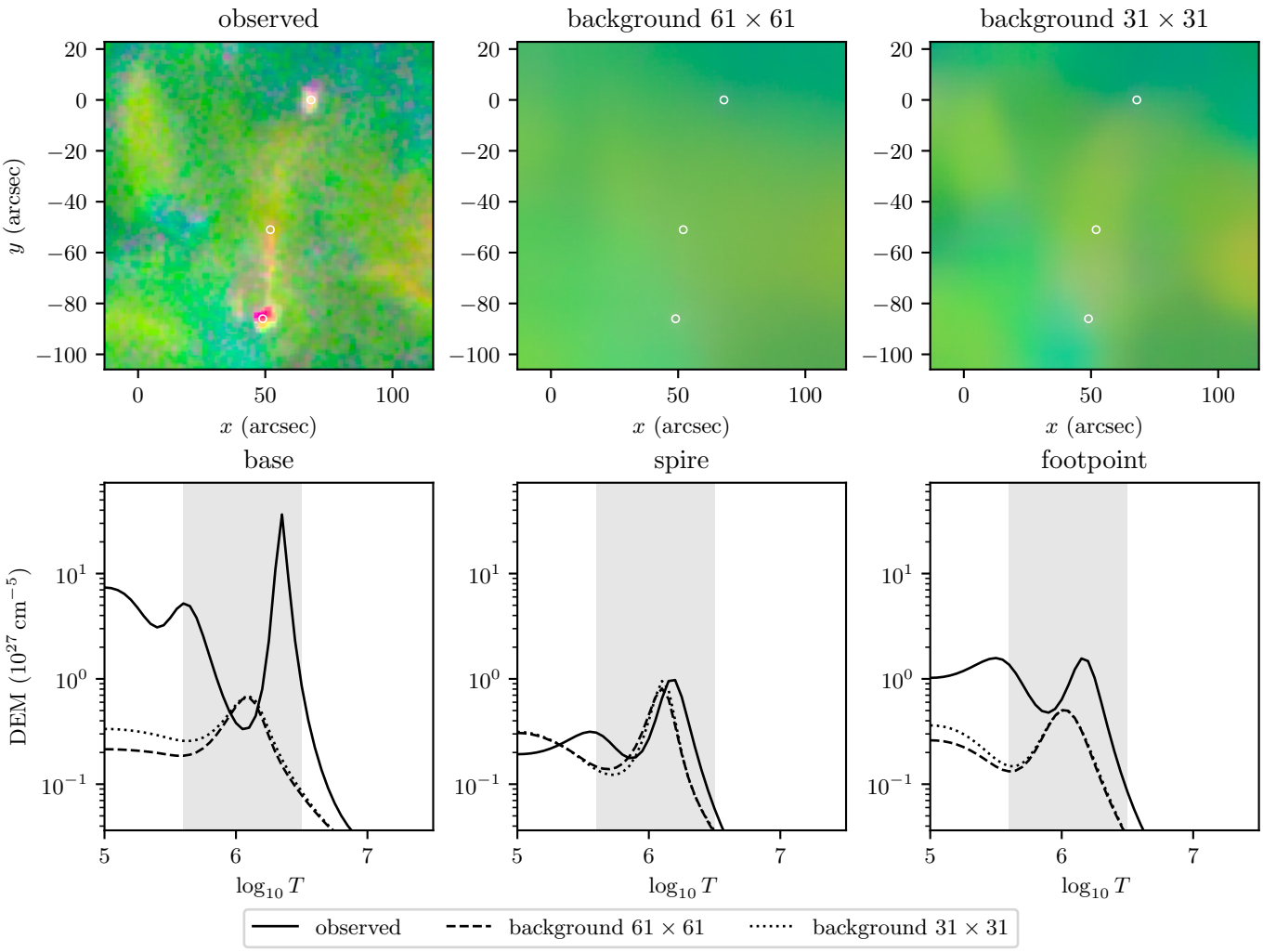


Figure 7. The DEM of event E at 18:08:00 UT, as in the top row of Figure 5 (left), and the DEMs inverted from its backgrounds of 61 by 61 (middle) and 31 by 31 pixels (right) in Figure 6, as false-color images colored as in Figure 5, on the brightness scale of its top row (top), and in the pixel nearest each of the three circled points (bottom), with the temperatures colored in the images shaded gray. White pixels are those where XRT saturated, which have no DEM.

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